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FinSage: A Multi-aspect RAG System for Financial Filings Question Answering

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Abstract

Leveraging large language models in real-world settings often entails a need to utilize domain-specific data and tools in order to follow the complex regulations that need to be followed for acceptable use. Within financial sectors, modern enterprises increasingly rely on Retrieval-Augmented Generation (RAG) systems to address complex information retrieval in financial document workflows. However, existing solutions struggle to account for the inherent heterogeneity of data (e.g., text, tables, diagrams) and evolving complexity in financial filings, leading to compromised accuracy in critical information extraction. We propose the FinSage framework as a solution, utilizing a multi-aspect RAG framework tailored for data retrieval and summarization in multi-modal financial documents. FinSage introduces three innovative components: (1) a multi-modal pre-processing pipeline that unifies diverse data formats and generates chunk-level metadata summaries, (2) a multi-path sparse-dense retrieval system augmented with query expansion (HyDE) and metadata-aware semantic search, and (3) a domain-specialized re-ranking module fine-tuned via Direct Preference Optimization to prioritize ground-truth-related content. Extensive experiments demonstrate that FinSage achieves an impressive recall of 92.51% on 75 expert-curated questions derived from surpasses the best baseline method on the FinanceBench question answering datasets by 24.06% in accuracy. Moreover, FinSage has been successfully deployed as financial question-answering system in online meetings, where it has already served more than 1,200 people. The implementation is publicly available at https://github.com/simplew4y/finsage.

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CCS Concepts

• **Information systems** → **Question answering**.

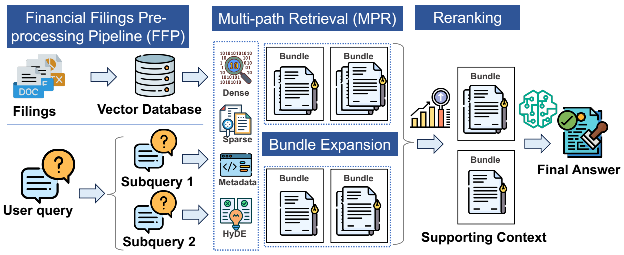

Figure 1: An overview of our proposed FinSage.

Keywords

LLM, Information Retrieval, Document Pre-Processing, Large Language Models, Retrieval Augmented Generation, Generative Models, Synthetic Datasets, Generative Retrieval

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1 Introduction

The integration of large language models into financial workflows represents both tremendous opportunity and significant operational complexity. Financial institutions operate within stringent database storage frameworks where the consequences of misinformation or oversight can result in substantial operational loss and reputation damage. This high-accuracy landscape creates unique demands for question-answering systems that must not only provide accurate information but also demonstrate verifiable adherence to evolving data complexity expectations [28, 29]. Financial question answering presents distinct challenges that differentiate it from general-domain applications. Unlike typical QA tasks that operate on homogeneous text corpora, answering financial questions requires processing complex multi-modal documents containing narrative disclosures, numerical tables, specific formatted for regulatory filings, and contextual relationships that span multiple filing periods, where the same data across different periods can be immensely different. The stakes are particularly high as errors in interpretation or misinformation may lead to inadequate risk assessment and reporting failures [31], hence causing catastrophic domino effect. Current retrieval-augmented generation (RAG) systems, while effective for general applications, encounter fundamental limitations when applied to financial QA scenarios. The heterogeneous nature of financial documents, which combines unstructured narrative sections, semi-structured tables, and contextual metadata—creates fragmentation issues that conventional text-centric retrieval architectures cannot adequately address [25, 36, 37, 41]. Traditional dense retrieval methods [17, 38] and sparse lexical matching approaches [26] often fail to capture the implicit and domain-specific concepts and terminologies critical for financial analysis [15, 32, 40]. Perhaps most critically, existing ranking mechanisms prioritize semantic similarity over domain relevancy, potentially surfacing contextually related but financial-domain-irrelevant information while overlooking operationally significant content [1, 3, 4]. This mismatch between retrieval objectives and operational expectations of financial institutions undermines the reliability needed for applications in financial industry, exposing financial institutions to operational and legal risks. To address these challenges, we introduce FinSage, a comprehensive financial question-answering framework specifically designed for complex financial filings summarization. Our approach recognizes that effective financial QA requires more than conventional retrieval—it demands a systematic integration of multi-modal processing, domain-aware retrieval strategies, and domain-relevancy-focused ranking mechanisms. As illustrated in Figure 2, FinSage consists of three synergistic components: (1) Firstly, the FFP module addresses multi-modal heterogeneity by employing specialized encoders for different content types while generating semantic metadata that preserves cross-modal relationships. (2) Secondly, the MPR component extends beyond traditional dense-sparse retrieval combinations by incorporating domain-aware query expansion and metadata-guided search strategies. (3) Finally, our re-ranking module learns to identify and prioritize passages containing domain-critical contexts and keywords, moving beyond generic semantic similarity to domain-specific relevance assessment. Through extensive evaluation, FinSage demonstrates substantial improvements over existing approaches, establishing a new benchmark for operational-oriented financial question answering. Our contributions encompass:

- We present an end-to-end financial QA system that integrates pre-processing, retrieval, and re-ranking to provide a robust solution for domain financial knowledge retrieval and summarization.
- We address key challenges in multi-modal financial document processing and domain-specific QA, bridging the gap between traditional RAG models and applications in complex database.
- We conduct extensive experiments and ablation studies to evaluate the effectiveness of our retrieval, re-ranking, and response generation mechanisms, analyzing the impact of different retrieval paths and re-ranking strategies.

2 Related Work

2.1 Retrieval Augmented Generation

Large language models (LLMs) are inherently limited by their fixed parametric size, affecting their ability to store and update factual knowledge. Retrieval-Augmented Generation (RAG) [8, 21] and retrieval-enhanced structure [6, 34] address this by integrating external knowledge through document retrieval. Recent advances include feedback mechanisms [2], self-referential knowledge [35], and domain-specific hierarchies [33]. Multi-head RAG approaches [4, 36] improve accuracy through diverse embeddings, while re-ranking methods [12, 14, 19] enhance precision by prioritizing relevant content.

2.2 Retrieval for Financial Data

Financial documents present unique challenges due to their multi-modal nature (text, tables, figures) and domain-specific terminology. While general multi-modal RAG systems [10, 25, 39] have shown

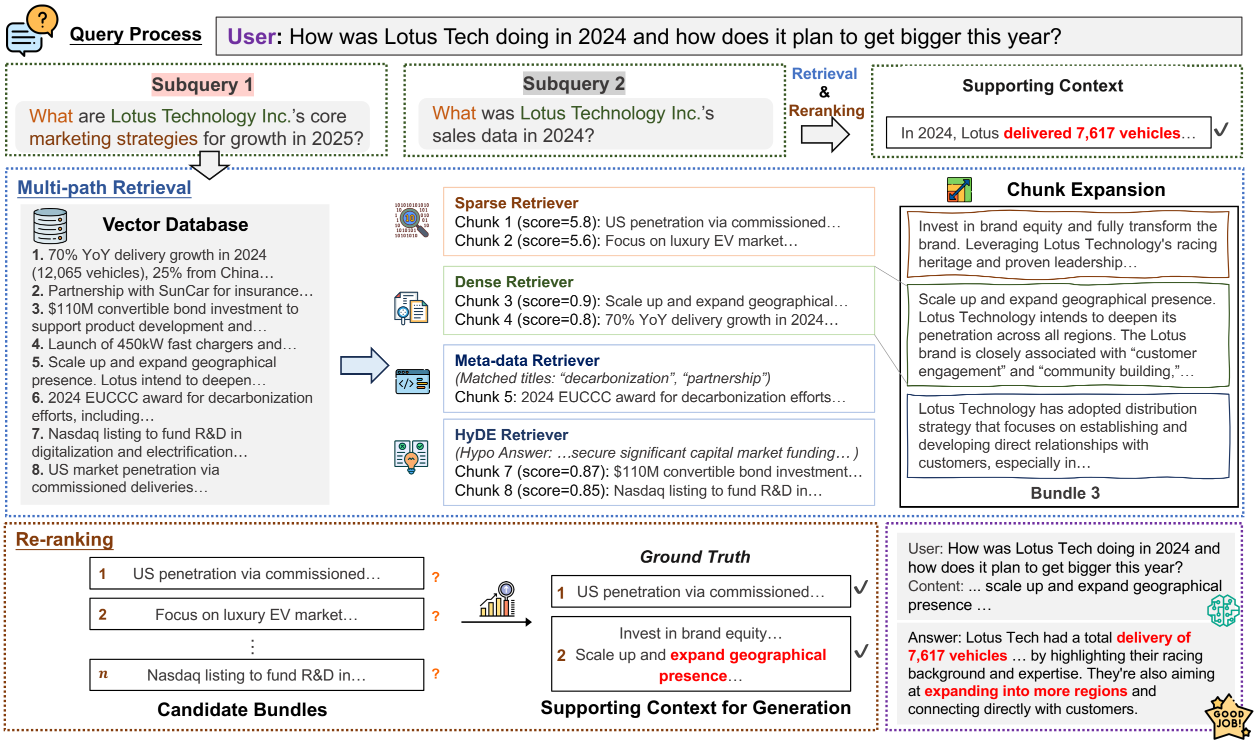

Figure 2: The complete FinSage RAG pipeline. The system processes the user query into subqueries. For each subquery, multiple chunks are retrieved with multi-path retrieval, and these chunks are then expanded into candidate bundles by concatenating neighboring chunks. After re-ranking, the most relevant chunks and their bundles are included for answer generation.

promise, financial applications require specialized approaches. Islam et al. [13] introduced FinanceBench for evaluating financial QA systems. Jimeno-Yepes et al. [15] explored chunking strategies for financial documents, and Setty et al. [30] proposed RAG pipelines for financial reports. However, existing methods struggle with complex multi-modal information and lack precision in domain-specific retrieval, motivating our comprehensive approach that addresses these limitations through specialized pre-processing, multi-path retrieval, and domain-adapted re-ranking.

3 Methodology

To reliably use RAG for financial document processing, the system must address three fundamental challenges: (1) handling multi-modal data such as tables, figures, and structured text; (2) extracting a comprehensive scope of relevant information from multiple sources; and (3) generating accurate and precise responses by effectively summarizing and reasoning over retrieved content. To tackle these challenges, we propose FinSage, an end-to-end financial QA system tailored for summarization of high accuracy and relevancy. This section introduces the three key components of FinSage, each designed to address specific challenges in financial RAG (Figure 2).

3.1 Financial Filings Pre-processing pipeline

We create a Financial Filings Pre-processing (FFP) pipeline to convert multi-modal financial filings into machine-readable text chunks. Figure 3 illustrates how FFP operates in two steps: textual encoding and semantic enhancement. Formally, let C denote the original set of multi-modal chunks extracted from a financial document, divided based on its chapters and a pre-defined token length. Each chunk $s_i = (t_i, m_i) \in C$ consists of raw content t_i and metadata m_i. The goal of the FFP is to transform chunks in C into $\tilde{C}$, a self-contained and condensed format that is better suited for downstream information retrieval tasks.

Textual Encoding of Multi-Modal Documents. The FFP pipeline first converts multi-modal financial documents into structured, machine-readable textual formats. Following the natural reading order, we use open-source PDF parsing tools [11] to decompose each document into a sequence of chunks. Each chunk is labeled with its corresponding type (i.e., text, figure, or table). Following Jimeno-Yepes et al. [15], we further transform figure and table chunks into descriptive textual narratives with large vision-language models. Specifically, figures are converted into structured captions summarizing their key insights, while tables are rewritten as textual statements capturing their primary data trends and relationships [42]. This transformation enables large language models (LLMs) to effectively interpret and utilize multi-modal content.

Semantic Enhancements. To improve the quality and coherence of the extracted texts, the FFP pipeline applies the following key steps.

(1) **Redundant chunk de-duplication.** We first de-duplicate documents to reduce redundancy, ensuring that duplicate content does not limit the diversity of retrieved information. For any

two chunks $s_i, s_j \in C$ such that $i \neq j$, we randomly discard s_i or s_j if the cosine similarity of the dense embeddings of their raw texts $\cos(\mathbf{d}_i, \mathbf{d}_j)$ exceeds a predefined threshold τ_{dedup}.

(2) **Co-reference resolution.** Next, we disambiguate the textual context of each chunk by iteratively resolving co-references. Within each subtitle section, pronouns of entity mentions are replaced with their corresponding antecedents identified from preceding chunks. Specifically, for each chunk containing pronouns, we employ an LLM as a co-reference resolver to perform entity mention replacements. The LLM receives k previous chunks to identify proper antecedents, where k is fixed.

(3) **Metadata generation.** Due to the token-length limitation, document sections may be separated into multiple chunks where content may no longer be self-contained. Thus, we ensure that chunks within a section retain sufficient context from others in the same section for retrieval. We employ an LLM to summarize each section of the financial filing and append the generated summary to the metadata of every chunk obtained from that section. This ensures that each chunk retains enough contextual information while maintaining a minimal context length, enhancing the efficiency and accuracy of document retrieval.

After the aforementioned process, for each chunk s_i, dense and sparse embeddings $\mathbf{d}_i$ and $\mathbf{p}_i$ are generated from $\tilde{t}_i$ using a text encoder and BM25, respectively. Furthermore, a dense embedding for the metadata $\mathbf{e}_i$ is also generated by a text encoder. The chunk is then represented as a 5-tuple $(\tilde{t}_i, \tilde{m}_i, \mathbf{d}_i, \mathbf{p}_i, \mathbf{e}_i)$. These enhancements ensure that extracted chunks are self-contained, contextually coherent, and semantically enriched, optimizing their utility for downstream processing.

3.2 Multi-path Retrieval pipeline

We introduce a Multi-path Retrieval (MPR) pipeline that enhances performance by leveraging semantic information. The pipeline consists of two modules: a Query Paraphrasing Module and a Retrieval Module for obtaining relevant answers from diverse sources.

Query Paraphrasing Module. An LLM reformulates each input query using FinSage's specialized vector store, which serves as a comprehensive knowledge base containing all company-related information. This paraphrasing process involves three key steps: (1) **Query Decomposition:** Each query is segmented into multiple self-contained sub-queries. This ensures that each subquery can be independently addressed by mapping its keywords to the corresponding segments in the knowledge base; (2) **Co-reference Resolution:** Co-references are resolved by converting the text into standardized English. In this process, all pronouns are replaced with their corresponding antecedents; (3) **Context Integration:** For queries referencing prior conversation history, relevant contexts are incorporated to maintain completeness and clarity.

Retrieval Module. To overcome limitations of traditional sparse (heuristic-dependent) and dense (resource-intensive fine-tuning) retrievers for domain-specific tasks, we develop a multi-path retrieval (MPR) pipeline including:

(1) **BM25 Sparse Retriever:** This component employs the BM25 scoring function to compute relevance based on tf-idf between the query and documents [26].

(2) **BGE-M3 Dense Retriever:** Utilizing the BGE-M3 model [9], this retriever calculates the cosine similarity between query embeddings and chunk embeddings. The retrieval process is accelerated by FAISS [16], ensuring high-performance search.

(3) **Metadata Retriever:** Leveraging the same BGE-M3 model, this module computes the cosine similarity between the query and metadata embeddings, facilitating the integration of structured metadata into the retrieval process.

(4) **HyDE Retriever:** This component computes the cosine similarity between hypothesis and chunk embeddings.

We collect all results selected by those retrievers into a chunk set, which serves as the output of the retrieval module.

Our metadata retrieval approach is designed to address high-level search tasks. We embed a shared summary (generated via FFP) as metadata for each document segment. Assigning this identical embedding to all chunks within a segment ensures they are retrieved together as a coherent group, strengthening contextual linkage.

The HyDE retriever is designed to address recall issues. In certain scenarios, target chunks and user queries exhibit weak correlations, even when the queries are optimally formulated. In such cases, the HyDE retriever leverages Hypothetical Document Embeddings[43], a specialized query expansion technique which uses an LLM-generated response to the user query as a hypothesis for retrieving similar chunks. To further enhance HyDE's performance, we augment the LLM's domain-specific knowledge through instruction fine-tuning on our custom Company dataset, following the InPars method [5]. Additionally, a smaller model is trained on a synthesized dataset to emulate the generative capabilities of larger models. As a result, HyDE not only improves the matching accuracy between the generated hypothetical responses and the target chunks but also bolsters overall retrieval performance across diverse content types.

Chunk Bundling Module. After retrieval, the MPR pipeline employs a chunk expansion mechanism to enhance contextual coherence among sequential chunks. This approach is particularly critical in financial documents, where key topics and discussions often span multiple chunks or paragraphs. Initially, each chunk is treated as an independent unit. These units are then iteratively expanded by evaluating adjacent chunks (preceding, following, or both) based on a predefined semantic similarity threshold.

3.3 Document Re-ranking

The Document Re-ranker (DRR) in FinSage acts as an intelligent filter, addressing shortcomings of initial retrieval to enhance accuracy, coherence, and reliability in final responses. The Document Re-ranker scores each candidate chunk $s_i \in \mathcal{R}_q$ with a Transformer-based decoder-only model.

Direct Preference Optimization. To further enhance ranking performance, we fine-tune the DRR using Direct Preference Optimization (DPO) [24], ensuring that our re-ranker effectively distinguishes and prioritizes the most relevant document chunks.

The DRR pipeline is iteratively implemented as follows:

(1) **Retrieval and Annotation**: Extensive results are retrieved using FAISS in the Multi-Path Retrieval pipeline (Section 3.2) for a rich set of candidate documents in each query. Retrieved documents are annotated to create preference samples for DPO,

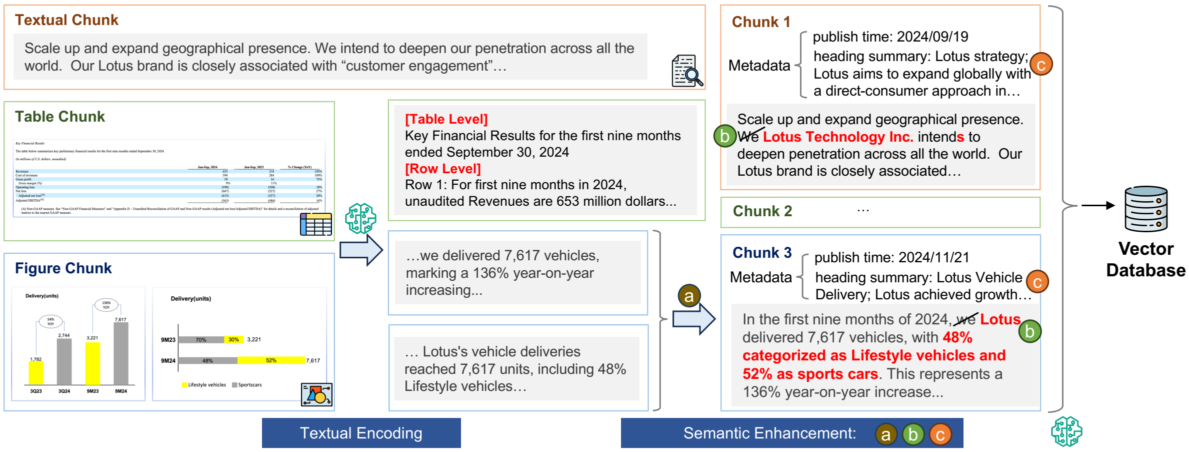

Figure 3: The FinSage FFP pipeline. Textual Encoding: PDF parsing tools extract multi-modal chunks (text, figures, tables), which are then converted into textual representations using large language models (LLMs). Semantic Enhancement: The textual representations are further refined by (a) eliminating redundant chunks through pairwise similarity comparison, (b) resolving co-reference within subtitle sections, and (c) generating subtitle-section-based summaries as metadata. These enriched chunks are then embedded in a vector database for further processing.

distinguishing between relevant and irrelevant matches. The annotated pairs are then organized into a structured dataset with the balance between positive and negative samples maintained to prevent training bias.

(2) **Evaluation**: Multiple metrics, including Normalized Discounted Cumulative Gain (NDCG) and Mean Reciprocal Rank (MRR) [27], are employed to verify the model's ability to determine document relevancy and hence improve DRR's performance.
(3) **Model Adjustment**: When the model fails to perform on a new test set, the retrieved documents are re-annotated to better represent the underlying patterns and relationships via a careful review and adjustment of training data. The model is retrained on top of the previous version using the updated annotations, refining the ranking ability. This is repeated until a satisfactory performance on the new test set is attained.

After document re-ranking, the RAG pipeline in FinSage follows the traditional method to handle user queries, retrieve related data from the vector database, and generate answers accordingly (Figure 2).

4 Experiments

We conduct experiments across three settings to evaluate FinSage: (1) assessing the Multi-path Retrieval pipeline, (2) evaluating the Document Re-ranker, and (3) measuring overall answer quality.

4.1 Datasets and Baseline

To evaluate performance, a manually annotated dataset is typically regarded as the "golden dataset," with high-quality entries serving as the ground truth [18]. We utilize two such datasets:

FinanceBench. FinanceBench [13] is a financial question-answering dataset created to evaluate the performance of LLMs on open-book tasks. We use 149 questions from FinanceBench and construct a single vector store by processing 83 documents that contain correct answers for these questions with FFP. A fixed length of 256 tokens is adopted when separating semantic segments into chunks. We compare our proposed RAG system with three baseline methods, namely Islam et al. [13], Jimeno-Yepes et al. [15] and Setty et al. [30] with their best-matching variants *Shared Vector Store*, *Base256*, and *Reranker*, respectively. These variants generally follow a setup similar to ours, utilizing a single vector store for all filings in FinanceBench, a chunk size of 256 tokens, as well as the retriever before generating answers. The baseline *reranker* is combined with a cross-encoder reranker which best aligns with our configuration.

In-depth Single Company Dataset. Apart from FinanceBench, we construct a custom QA dataset using 12 filings from a single company. These filings are segmented into a vector database, with each chunk fixed at 200 tokens. We manually compile 75 question-answer pairs from SEC 6-K and F-1 filings, associating each question with an average of 8.7 relevant text chunks containing the most pertinent passages. The questions cover various aspects of Lotus Technology Inc., including products, sales data, history, and corporate strategy. To ensure accuracy, we engage domain experts to review the manually annotated chunks, establishing them as the ground truth. The complexity of these questions often necessitates aggregating information from multiple documents or sections, making it challenging to locate answers within a single chunk. We refer to this dataset henceforth as the Company dataset.

4.2 Experiment Settings

4.2.1 Multi-path Retrieval (MPR) settings. `Qwen-2.5-7B-Instruct` is our HyDE model, which we train as follows: question-document pairs are first generated using this model architecture, and the top 2,000 are selected by similarity. Each selected pair then yields two hypothetical chunks. During inference, the trained model generates three hypothetical answers per query to retrieve top-K chunks.
For metadata retrieval, we select top-K metadata entries and include all chunks from corresponding semantic segments. We set chunk

Table 1: Retrieval Performance on the company dataset. "Avg Num Retrieved" represents the average number of chunks in the retrieved set. Recall, Precision, and F1 scores are calculated based on the relevant chunks retrieved.

Category	Method	Avg Recall	Avg Precision	Avg F1	Avg Num Retrieved
FAISS Retrieval	FAISS (Baseline)	0.8194	0.0941	0.1646	75.6
	+ Bundle Expansion (Exp)	0.8103	0.0999	0.1733	70.25
BM25 Retrieval	+ BM25	0.8452	0.1147	0.1949	66.24
Metadata Retrieval	+ Metadata	0.8573	0.1198	0.2037	64.87
HyDE Retrieval	HyDE-1: HyDE(Qwen7B)	0.8228	0.1076	0.1830	69.09
	HyDE-2: HyDE(Qwen7B-SFT)	0.8567	0.1072	0.1831	72.92
	HyDE-3: HyDE(Qwen72B)	0.8323	0.1078	0.1844	69.77
FinSage	+ BM25+Metadata+HyDE-2 + Exp	**0.9251**	**0.1272**	**0.2156**	68.75

expansion threshold $\tau_{\text{exp}} = 0.85$ with limit $\mathcal{K} = 5$ to include the neighbor chunk with sufficient similarity.

4.2.2 Document Re-ranker (DRR) settings. For re-ranking, we compare **fine-tuned DRR**, trained on single-company financial filings, against `BAAI/bge-reranker-v2-gemma`, a general re-ranker. Our model is fine-tuned using DPO with a contrastive learning objective. The training dataset comprises annotated query-document pairs, each containing one relevant chunk and multiple irrelevant ones.

4.2.3 Response settings. We compare FinSage's QA performance against Jimeno-Yepes et al. [15], Setty et al. [30] and Islam et al. [13] on FinanceBench [13]. For FinanceBench evaluation, we construct a single vector store (chunk size = 256) from financial PDFs using FFP, then apply multi-path retrieval with our fine-tuned re-ranker to retrieve relevant chunks before answering, which resembles the setup of our baselines. For the Company dataset, we evaluate accuracy against ground-truth. For response generation, we set K = 5 for each sub-query to efficient token usage, following empirical evidence that five chunks suffice for effective answers [20].

4.3 Evaluation

MPR Evaluation. Our MPR pipeline is evaluated using various combinations of retriever paths by analyzing document chunk recommendation metrics (Recall, Precision, F1) on the in-depth dataset. By introducing additional retriever paths, we expect improvements in recall rates due to a more diverse information coverage [45].

DRR Evaluation. The Document Re-ranker's (DRR) effectiveness is assessed by comparing its retrieved chunks against the manually annotated ground truth dataset. The DRR selects up to K top chunks per sub-query (K is adjustable). Performance is measured using Precision, Normalized Recall, Mean Reciprocal Rank (MRR), and binary Normalized Discounted Cumulative Gain (nDCG). For fair assessment, we use an adjusted **Normalized Recall** = num. of correctly retrieved chunks / min(ground truth chunks, K). This denominator cap prevents unfairly penalizing optimal re-ranking within the K-chunk retrieval limit.

Response Evaluation. FinSage's response quality is automatically evaluated against expert answers using `GPT-4o` via LlamaIndex's *CorrectnessEvaluator* module, which scores alignment with ground truth. This LLM-as-judge approach aligns with prior work [22, 23, 44]. Additionally, our team manually evaluated responses for factual correctness, coherence, and completeness, tolerating minor non-critical inconsistencies. Further details and resources are on

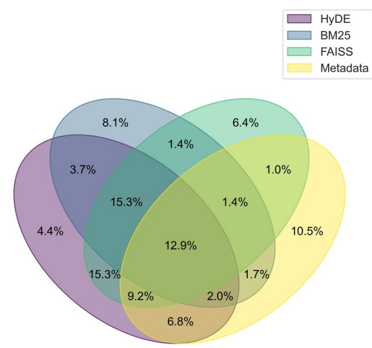

Figure 4: Percentage of relevant chunks selected relative to the total number of retrieved relevant chunks.

our GitHub repository. These combined methods assess FinSage's textual and semantic alignment with ground truth answers.

5 Results and Analysis

Table 2: Performance of End-To-End QA. Our implementation achieves the top LLM and Manual scores.

Dataset	Method	LLM	Manual
FinanceBench	Islam et al. [13]	-	0.1900
FinanceBench	Jimeno-Yepes et al. [15]	0.3262	0.3688
FinanceBench	Setty et al. [30]	0.2560	-
FinanceBench	FinSage*	**0.4966**	**0.5705**
Company	FinSage*	0.8533	0.8800

5.1 MPR Results

Table 1 demonstrates FinSage's superior performance through the multi-path retrieval architecture, particularly its integration of metadata-aware retrieval with fine-tuned HyDE generation, allows it to comprehensively capture both the structural and semantic aspects of financial documents. Experimental results show our fine-tuned `Qwen2.5-7B-SFT` model outperforms the much larger `Qwen2.5-72B`, particularly in recall in average. This underscores the critical role of fine-tuning with high-quality, domain-specific data and supports findings that well-adapted smaller models can rival larger ones in specialized tasks while offering significant computational advantages [7].

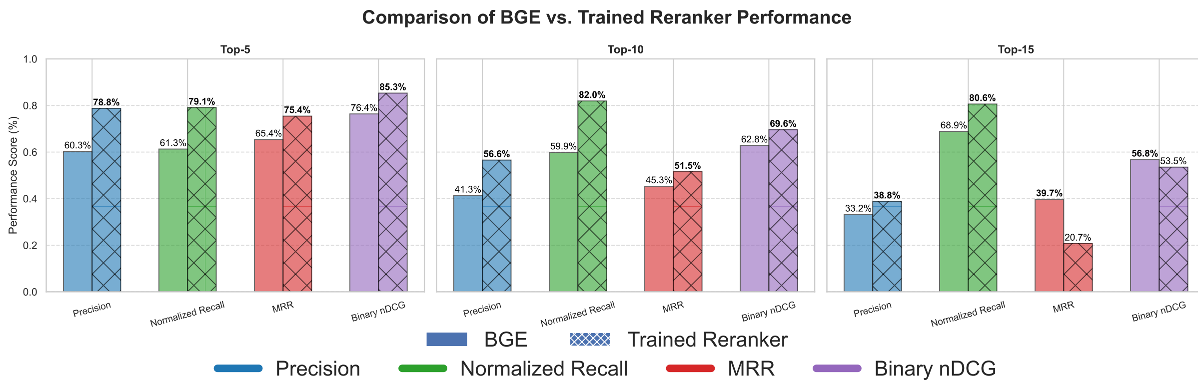

Figure 5: Performance of different re-rankers under different retrieval settings.

To better understand the uniqueness of each retrieval method, we analyze their overlap on the Company dataset. As shown in Figure 4, while each method uniquely contributes to retrieving positive chunks, there is significant overlap. Notably, FAISS and HyDE retrieve highly similar positive chunks, with an overlap rate of 52.7% (15.3%+12.9%+15.3%+9.2%), compared to their individual recovery rates of 62.9% and 69.6%. This overlap likely arises because HyDE generates hypothetical documents by referencing the original document, preserving much of its context.

5.2 DRR Results

We compare re-rankers' performance in Figure 5. The results demonstrate that our Document Re-ranker significantly and consistently outperforms `bge-reranker-v2-Gemma` across all settings and metrics in top-5 and top-10 configurations. Notably, our re-ranker improves recall by 15% across all setups and substantially boosts precision from 5.6% to 18.5%. It is worth noting that precision, MRR, and binary nDCG all decline as more chunks are extracted by the reranker: precision drops from 78.8% to 38.8%, MRR from 75.4% to 20.7%, and binary nDCG from 85.3% to 53.5%. This consistent trend across benchmark scenarios indicates that extracting additional chunks increases the burden on the re-ranker, resulting in less optimal ranking. Moreover, the rapid degradation in these metrics between Top-10 and Top-15 aligns with previous studies suggesting that retrieving only 5 to 10 chunks is generally sufficient for producing effective answers [20].

5.3 Response Results

Table 2 shows performance on the FinanceBench and Company datasets. FinSage achieves a significant improvement over prior methods on FinanceBench, with an LLM accuracy of 49.66%, outperforming Jimeno-Yepes et al. [15] (32.62%) and Setty et al. [30] (25.60%, aligns most closely with our configuration). Similarly, manual evaluation reflects an accuracy of 57.05%, substantially higher than the closest competitor Jimeno-Yepes et al. [15] (36.88%). FinSage also excels on the Company dataset, which reaches accuracies of 85.33% (LLM) and 88.00% (mannual). This demonstrates the FinSage's strategy of retrieving and ranking semantically relevant passages from structured documents significantly improves answer quality. In terms of cost and efficiency (Table 3), an average request takes approximately 15 seconds to generate the first token and costs about \$0.03 using gpt-4o, with computational overhead stemming from LLM and re-ranker inference.

Table 3: System Efficiency and Deployment Cost. n denotes the number of subqueries

Step	Process	Time (s)	Token Usage	Cost Estimation
1	Query rewriting	~2.5	~1200	~\$0.005
2	HyDE	~4.2	~500	~\$0.002 × n
3	Retrieval & reranking	~4.7	N/A	Local
4	Answering	~4.7	~2500	~\$0.012 × n
5	Answer merging	~1.7	~200 + 200 × n	~\$0.002 + \$0.002 × n
Total	–	**~13–17**	~3.7k + 3k × n	**~\$0.017 + \$0.016 × n**

Table 4: Faithful Evaluation Score Comparison.

Method	Questions	Mean	Median	Min	Max	Pass %
GraphRAG	71(w/ 4 failures)	3.46	3.50	2.0	5.0	42.50%
LightRAG	75	2.45	2.00	1.0	5.0	13.67%
FinSage	75	4.31	5.00	1.00	5.00	**82.67%**

5.4 Comparison with Other RAG Solutions

Our method is also compared with mainstream graph-based RAG solutions **GraphRAG** and **LightRAG** Table 4 highlights Finsage's advantages in response time and performance. This suggests graph-based complexity offers no quality improvement in our experimental context, underscoring the importance of balancing speed and performance for domain-specific RAG implementations. Detailed settings and analysis are available on our GitHub repository.

6 Conclusion

In this paper, we propose FinSage, a RAG solution for the question-answering task on financial filings. FinSage comprises a multi-modal data pre-processing pipeline, a domain-aware multi-path retrieval system specifically designed for financial data, and a domain-specialized re-ranking module fine-tuned using DPO. Our system demonstrated state-of-the-art performance on expert-curated questions derived from financial filings. Moreover, FinSage has been deployed as a financial question-answering system in online meetings organized by affiliated entrepreneurs.

Usage of Generative AI

We make use of ChatGPT to check potential grammar mistakes and refine our writing. We adopt Qwen2.5 and GPT4o to conduct our experiments and make judgments.

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